

NLP-to-SQL Translator

Fine-Tuning & Schema Grounding for CodeGen-350M

Capstone Project | Implementation Phase Progress

Executive Summary

Evolution from zero-shot baselines to optimized grounded generation.

 Abstract AI Network

Mission & Baseline Establishing



Primary Goal

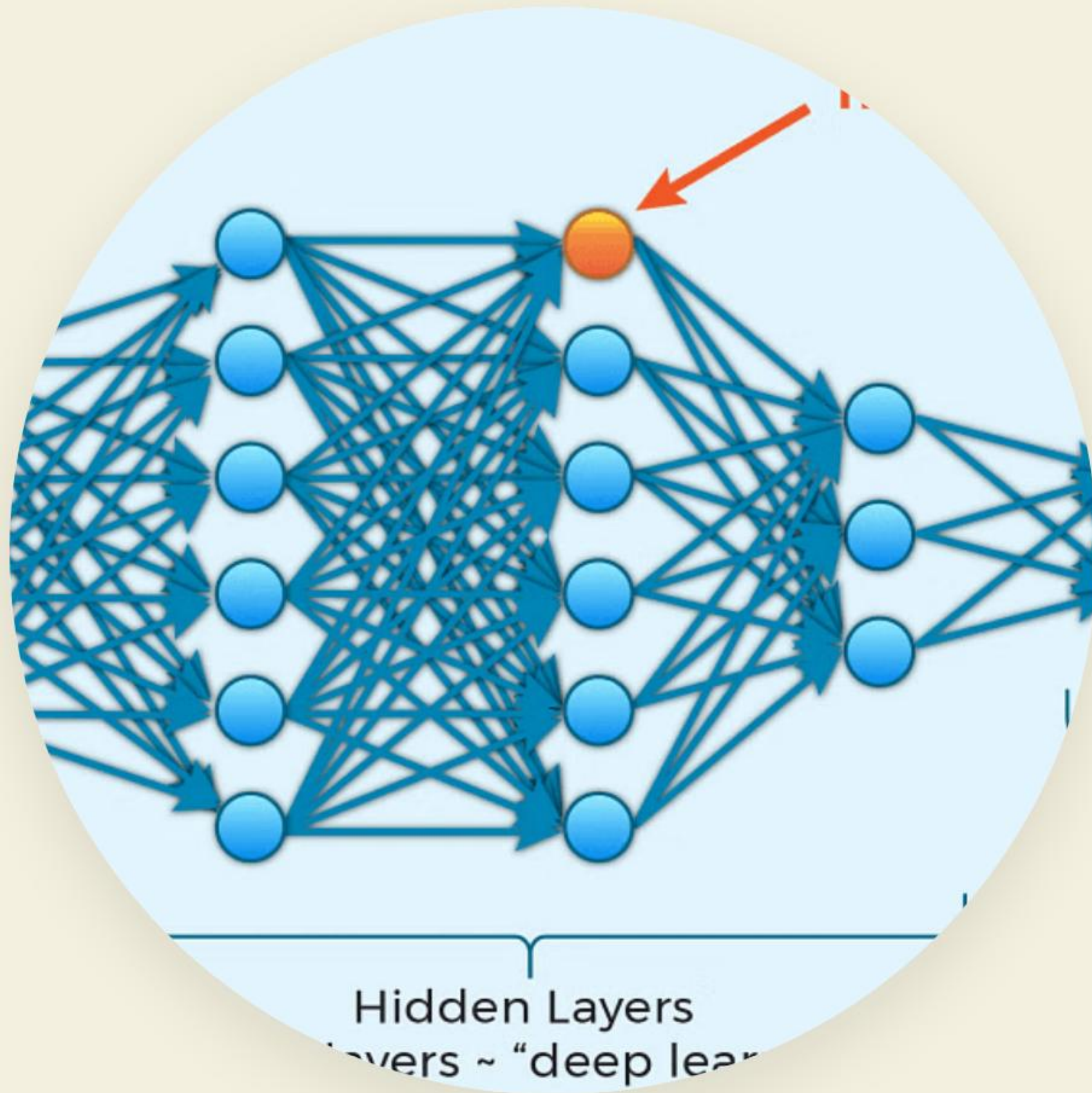
Develop a robust Natural Language interface for Relational Databases using small, parameter-efficient language models.



The Baseline

Establishing zero-shot performance "cliffs." Confirmed near-zero scores on MongoDB, necessitating fine-tuning for specific DSLs.

Core Model Architecture



Salesforce CodeGen-350M

Variant: Mono (Python/Code optimized)

Parameters: 350 Million

License: Apache 2.0 (Open Source)

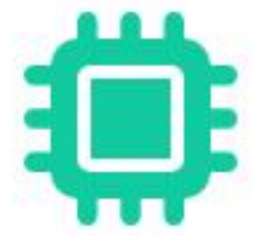
Rationale: Lightweight enough for laptop inference while maintaining competitive syntax understanding for program synthesis.

Spider Benchmark Dataset

Category	Details	Purpose
Validation Tasks	1,034 Entries	Final performance reporting
Databases	200 Relational DBs	Cross-domain generalization test
Schema Types	SQLite / MySQL	Real-world DDL consistency

Spider requires zero-shot schema generalization, meaning the model is tested on databases it never saw during training.

PEFT & LoRA Fine-Tuning



Efficiency

Used LoRA (Low-Rank Adaptation) to freeze base weights and only train small adapter modules.



Integration

Leveraged Hugging Face PEFT library to inject adapters into the attention layers.



Hardware

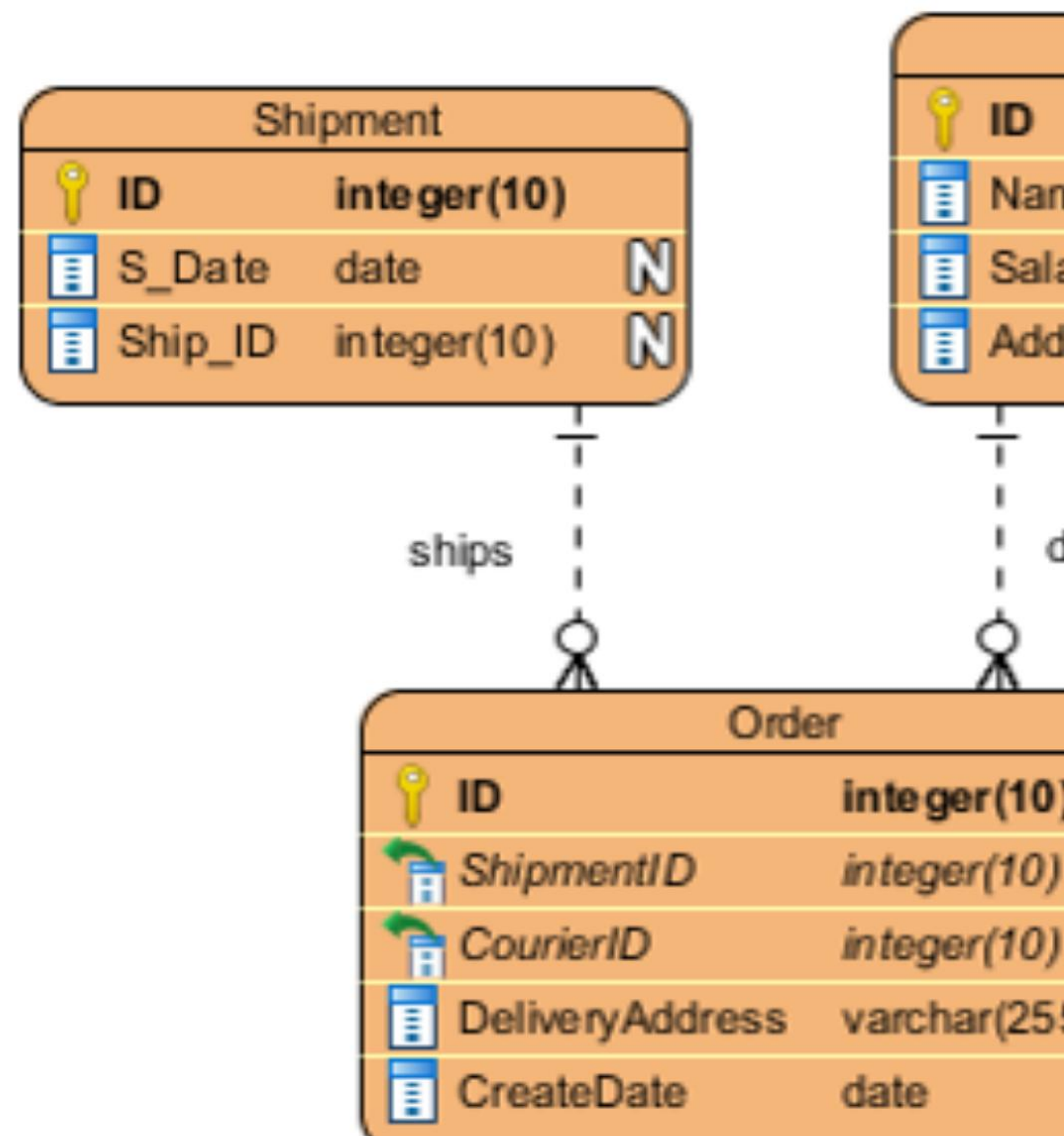
Balanced 8-bit quantization to enable training on consumer-grade hardware configurations.

Stage 3: Schema Grounding

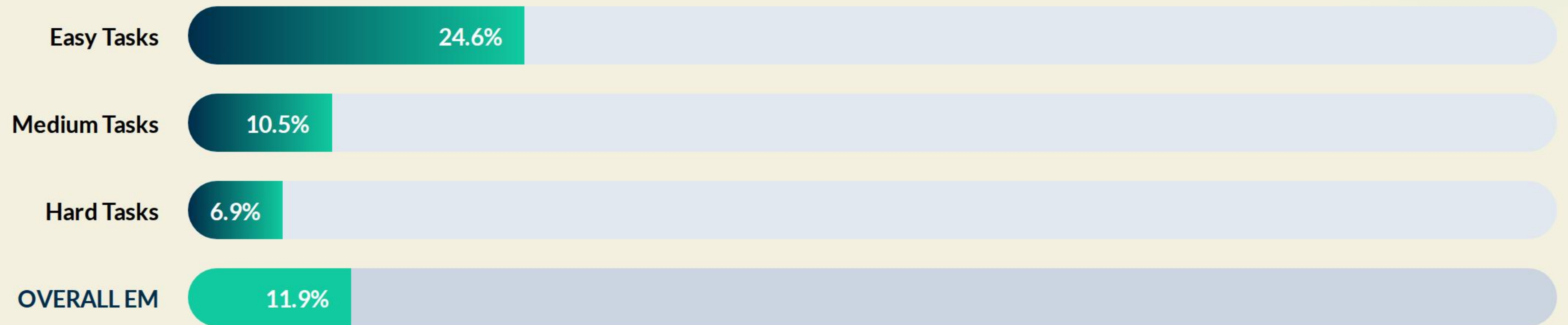
To resolve column hallucination, we implemented **Dynamic DDL Injection**.

Every prompt includes the structural metadata of the target database: Table X [Col1, Col2...].

This forces the model to "look" at the available catalog before generating the query, bridging the gap between language and data architecture.



Exact Match Accuracy Breakdown



Performance reflects the 350M parameter capacity on complex out-of-distribution databases.

Precision Insight: Select Clause

92.3%

HARD TIER PRECISION

Mapping Success

While Exact Match is stringent, partial matching reveals the model's true capability.

On "Hard" difficulty tasks, the model identifies the correct **SELECT** columns with **92.3% precision**.

This proves that **Schema Grounding** effectively solves column-level hallucinations.

Error Taxonomy & Limitations

The IUEN Gap (0.0%)

Model fails consistently on Intersect, Union, Except, and Nested queries.

Reason: Lack of parameter capacity to track multi-step relational dependencies.

Low F1 on Joins

Complex multi-table joins trigger logic drift.

Reason: Attention fragmentation in 350M models when DDL context grows too large.

Operational Transition

Local CPU



Initial code testing and dataset parsing.

Colab T4



Full scale generation for 1,034 validation tasks.

GPU Accel.



Optimized 16-bit inference for speed.

Analysis



Local metric parsing and report generation.



Next Steps

Expansion to SQL-to-NoSQL and capacity scaling.

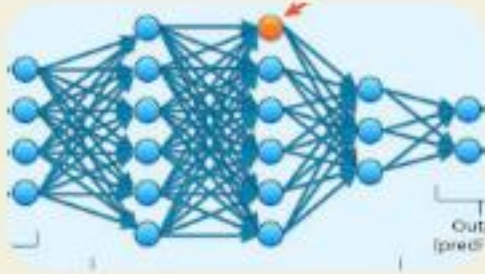
Thank you! | Questions?

Image Sources



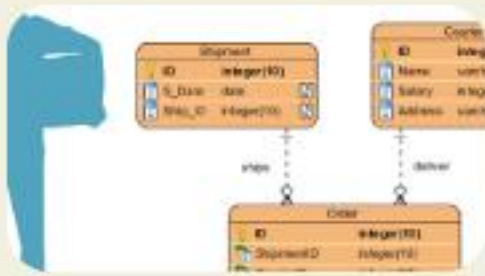
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Source: easy-peasy.ai



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